ELEC 576 Final Project Proposal

Automated Document Classification for Academic Research

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Abstract

This project aims to develop a deep learning model for automatically classifying academic research papers by their primary field of study. Leveraging convolutional neural networks (CNNs) and natural language processing (NLP) techniques, this model will analyze the content and structure of research papers to predict fields like computer science, biology, and physics. The expected impact includes streamlining literature reviews, facilitating interdisciplinary research, and aiding academic search engines.

Background/Motivation

Research papers are published at an accelerating pace, and researchers often spend significant time sorting and identifying relevant literature. Traditional keyword-based search engines sometimes misclassify documents or fail to capture nuanced subject boundaries, making literature reviews time-consuming. Current work primarily focuses on general text classification without much consideration for the academic context. With an automated classification system, this project proposes a novel application of CNNs and NLP in academia to improve research workflows by categorizing documents based on content and style, rather than keyword occurrence alone.

Proposed Experimentation/Implementation

- **Dataset:** The dataset will consist of publicly available research papers from platforms like arXiv. Papers will be categorized based on arXiv metadata, providing a labeled dataset for training and testing.
- **Preprocessing:** Text preprocessing will involve tokenization, stopword

removal, and vectorization (possibly using word embeddings or TF-IDF). For CNN inputs, sentences or abstracts will be converted into sequences of word embeddings.

- **Architecture:** The model will begin with a simple CNN for text classification, with plans to experiment with LSTM or Transformer-based layers if time permits. Hyperparameters will be optimized based on validation performance.
- **Implementation Steps:**
    1. Dataset gathering and cleaning
    2. Model architecture design and baseline implementation
    3. Training, tuning, and testing
    4. Evaluation using metrics like F1-score and accuracy
- **Evaluation:** Success will be measured by the model's classification accuracy across the most common fields (expected to be ~70-80% for a top-three accuracy).

Feasibility and Limitation

The primary limitation is time, as training and fine-tuning more complex architectures like Transformers might exceed a single researcher's capacity. Additionally, class imbalance among research fields could skew results, which may require techniques like data augmentation or weighted loss functions to counterbalance.

Potential Impact

Automated classification could significantly reduce time spent on literature searches, support interdisciplinary research by identifying relevant papers in related fields, and enhance the accuracy of academic search engines. Over time, this tool could be adapted to prioritize high-impact papers, facilitate grant proposal preparations, and even predict trending research topics.

References

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[3] arXiv Dataset: https://www.kaggle.com/datasets/Cornell-University/arxiv/data